ay/date

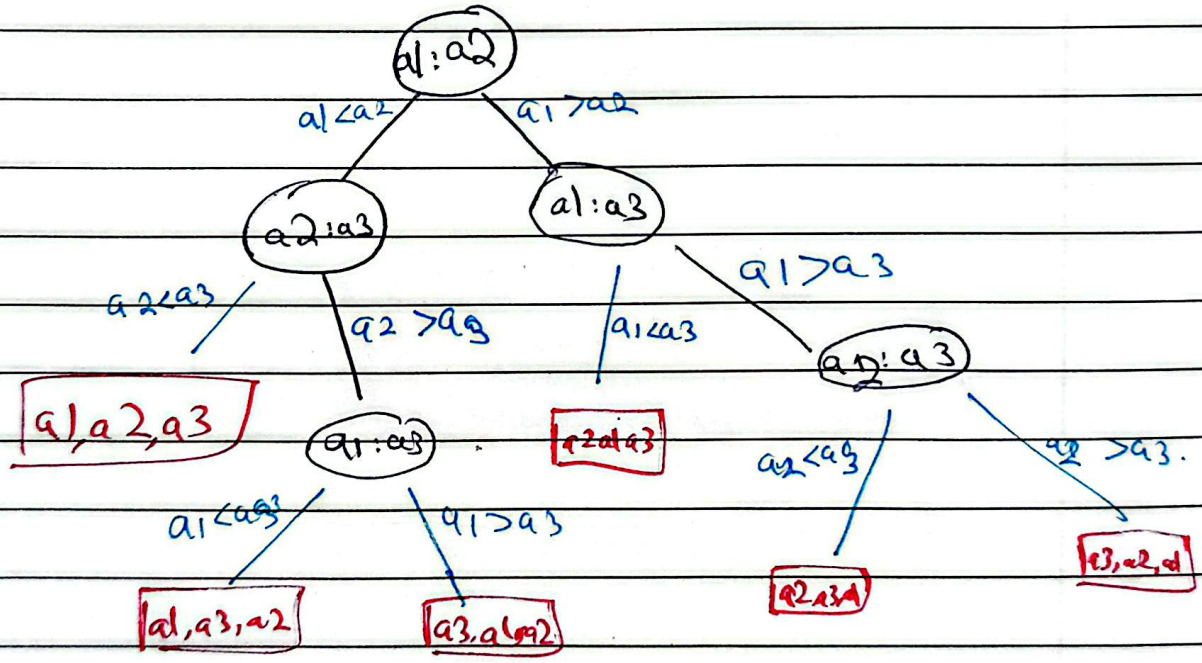
Comparison based Sorts:

Till now all sorts we were doing were comparison based sorts.

In all these we saw that our best possible worst case way $n \log n$.

but can't we do any better?

Consider an array of 3 elements: $\langle a_1, a_2, a_3 \rangle$
(we are using a decision tree).



□ sorted.

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So if we see these comparisons we get to know that for 3 elements there are 3! posible results.

so total amount of leaves = n!

& we know that it is a tree & so the log from root to tree:

log (n).

so

or complexity = log (n!)

and as log (n!) ≤ n log n.

so best posible worst case:

$\Theta(n \log n)$.

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